Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/06/2015
PI Name:	Bhattacharya, Sharmila		
Project Title:	Gravity and oxidative stress responses in Drosophila melanogaster		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Organization Name:	NASA Ames Research Center		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	03/01/2010	**End Date:**	02/28/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 2/28/2014 per A. Chu/ARC (Ed., 3/4/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space environment of microgravity and radiation imposes stress conditions that affect the cellular and molecular responses of organisms and dictate their survival. A very fast response to stress is the production of reactive oxygen species (ROS), followed by intracellular oxidation. Space flight is associated with increased oxidative damage after return to 1 g, suggesting that microgravity and/or the transition between gravity fields triggers a response. We propose to use a model organism used previously for space flight, the fruit fly Drosophila melanogaster. Studies in both hyper- and microgravity have already established many of the physiological alterations, and work from our own laboratory suggests that oxidative stress is implicated in the molecular response under both conditions. We will use hypergravity as a test environment for gravity alterations, and to dissect out molecular and cellular responses as well as long-term physiological outcomes. The goal is to provide a complete picture of the oxidative stress pathway in hypergravity, and to determine and select the key molecular functions and physiological read-outs to be tested in microgravity. The aims are to quantify gravity-dependent cellular ROS production and to determine the key points of entry into the oxidative pathway by using mutants that show a physiologically altered response to hypergravity. This approach will give a picture of the stress pathway, starting from fast cellular changes triggered by sudden changes in the gravity vector, to tissue-specific gene and enzyme activity alterations. Understanding the stress pathway in hypergravity opens the venue into looking at the opposite transition in the gravity load, from 1 g to microgravity during space flight, and will help predict the cellular and physiological response to other stresses imposed by the space environment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Some of the imaging tools developed to measure the behavioral changes for this project will be useful for a variety of behavioral studies in different organisms on Earth. The general issue of oxidative stress is also of great relevance to aging and stress on Earth, and the studies here therefore provide yet another paradigm in which to study and understand these molecular responses.
Task Progress:	This work has proved our starting hypothesis that hypergravity is linked closely with alterations in oxidative stress mechanisms and that these can also influence behavioral responses to gravity. By using a genetically tractable organism like the fruit fly to identify the specific molecular pathways involved in the response to altered gravity conditions, we have been able to identify specific genes and specific anatomical locations within the fly brain that are important for these responses to gravity. By using the hypergravity environment to identify important genetic determinants of physiological changes in the fly, we hope to extend this work in the future to the microgravity environment of space flight and test different ways in which to make the organism more resistant to the changes induced by the space flight environment. Two manuscripts are in preparation for submission to peer-reviewed journals with expected publication later in 2015: Ravikumar Hosamani, Oana Marcu, and Sharmila Bhattacharya "Hypergravity-induced altered behaviour in Drosophila" Manuscript in preparation. Ravikumar Hosamani and Sharmila Bhattacharya "Acute Hypergravity Causes Altered Gene Expression, and Affects the Oxidative Stress Pathway in Drosophila" Manuscript in preparation.
Bibliography Type:	Description: (Last Updated: 03/02/2021)
Abstracts for Journals and Proceedings	Gegenhuber JB, Hosamani R, Bhattacharya S. "Chronic Hypergravity Affects Oxidative Stress Pathways in Drosophila melanogaster." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. (Oral Presentation/talk) 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Hosamani R, Leib R, Bhattacharya S. "The Effect of Altered Gravity on the Drosophila Proteome: Relevance to NASA's Space Flight Mission." Mass Spectrometry Users' Conference, Stanford University, Stanford, CA, Oct 7, 2014. (Invited Talk) Mass Spectrometry Users' Conference, Stanford University, Stanford, CA, Oct 7, 2014. , Oct-2014
Abstracts for Journals and Proceedings	Hosamani R, Adams C, Bhardwaj SR, Okumu A, Chien AS, Bhattacharya S. "Proteome expression profiling of hypergravity exposure in Drosophila; preparing for NASA space mission." 62nd ASMS Conference on Mass Spectrometry and Allied Topics Baltimore, Maryland, June 15-19, 2014. (Oral presentation/talk) 62nd ASMS Conference on Mass Spectrometry and Allied Topics Baltimore, Maryland, June 15-19, 2014. , Jun-2014
Abstracts for Journals and Proceedings	Lera M, Hosamani R, Bhattacharya S. "Fruit Fly Lab: A Hardware Suite with Flight Heritage Developed to Support Drosophila Research Onboard ISS." 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. (Oral presentation/talk) 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. , Jun-2014
Abstracts for Journals and Proceedings	Hosamani R, Weinrich J, Reddy C, Bhattacharya S. "Acute hypergravity alters locomotor behavior and causes physiological stress in Drosophila melanogaster." 55th Annual Drosophila Research Conference, San Diego, CA, March 26-30, 2014. 55th Annual Drosophila Research Conference, San Diego, CA, March 26-30, 2014. Abstract 523A. http://www.genetics-gsa.org/drosophila/2014/pdf/Abstracts%20Full.pdf ; accessed 1/7/2015. , Mar-2014
Abstracts for Journals and Proceedings	Bhattacharya S, Lera M, Gresser A, Hosamani R. "Drosophila Habitat Developed to Support Research On-board the International Space Station." 55th Annual Drosophila Research Conference, San Diego, CA, March 26-30, 2014. 55th Annual Drosophila Research Conference, San Diego, CA, March 26-30, 2014. , Mar-2014

Abstracts for Journals and Proceedings	Hosamani R, Bhattacharya S. "Acute Hypergravity Causes Altered Gene Expression, and Affects the Oxidative Stress Pathway in Drosophila." 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. 29th American Society for Gravitational and Space Research and 5th International Symposium for Physical Sciences in Space, Orlando, FL, November 3-8, 2013. Program and abstracts, p. 89. https://asgsr.org/images/stories/pdf/Complete_Program_FINAL_2013_Meeting.pdf ; accessed 1/7/2015. , Nov-2013
Articles in Peer-reviewed Journals	Hateley S, Hosamani R, Bhardwaj SR, Pachter L, Bhattacharya S. "Transcriptomic response of Drosophila melanogaster pupae developed in hypergravity." Genomics. 2016 Oct;108(3-4):158-67. Epub 2016 Sep 10. https://doi.org/10.1016/j.ygeno.2016.09.002 ; PubMed PMID: 27621057 , Oct-2016
Articles in Peer-reviewed Journals	Parsons-Wingerter P, Hosamani R, Vickerman MB, Bhattacharya S. "Mapping by VESGEN of wing vein phenotype in Drosophila for quantifying adaptations to space environments." Gravit Space Res. 2015 Dec;3(2):54-64. http://gravitationalandspacebiology.org/index.php/journal/article/view/693 , Dec-2015